

Computational Approaches to Exploring Transcription Factor-DNA Interactions: From Resource Development to Co-binding Prediction

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Abstract

Transcription factors (TFs) play a crucial role in regulating the transcription of DNA into RNA. TFs bind to regulatory DNA regions, ensuring that specific genes are transcribed at the right time and place. The main targets of TFs are cis-regulatory regions (CRRs), known as promoters and enhancers. Promoters are located around transcription start sites (TSSs), while enhancers can be found at large linear distances from the genes they regulate. TFs bind promoters and enhancers to recruit other proteins and RNA polymerase II to initiate the transcription. TFs are sequence-specific; their DNA-binding domains (DBDs) interact with short DNA nucleotide sequences called transcription factor binding sites (TFBSs). Understanding gene regulatory mechanisms requires investigating how TFs interact with DNA and identifying the patterns they recognize.

While the binding properties of individual TFs are well-documented, transcription regulation is complex and often involves multiple TFs working together. Different combinations of TFs can interact with the DNA cooperatively by co-binding the same region or forming protein complexes. The distribution of TFBSs within CRRs determines the binding of different TFs and can have a different regulatory effect. The orientation and spacing in between TFBSs is known as DNA motif grammar. Many cellular processes, such as cell differentiation and response to external stimuli, rely on transcription regulation by cooperating TFs. Studying these combinatorial TF bindings is challenging but essential for understanding the full complexity of transcription regulation.

Experimental techniques like chromatin immunoprecipitation followed by sequencing (ChIP-seq) can determine the genomic sequences bound by a TF of interest *in vivo*. Computational methods then model TF-DNA interactions. The most widely used computational model for TF-DNA interactions is the position weight matrix (PWM). PWMs summarize experimentally derived sequences bound by a TF and define a DNA motif recognized by a particular TF. Motifs can be discovered *de novo* from a set of genomic sequences by identifying overrepresented patterns. Computational models of TF-DNA interactions are stored in databases and can be used to predict TFBSs, annotate unknown motifs, or predict TF co-binding events. JASPAR is an example of a database that contains collections of TF binding motifs for TFs from different taxonomic groups.

The main goal of this thesis was to employ computational approaches to explore TF-DNA interactions by (1) discovering new TF binding motifs and updating the JASPAR database while also developing and updating robust pipelines for automated data processing and post-processing, and (2) identifying and characterizing co-binding DNA patterns.

The first objective was achieved by contributing to the two latest releases of the JASPAR database. For both releases, a vast amount of publicly available genomic data was collected to discover DNA binding motifs. These motifs were then manually curated and added to the CORE and UNVALIDATED collections, depending on the available orthogonal support. In addition to adding new information to the database, existing profiles were revised to ensure metadata quality. The final sets of motifs were used for downstream analyses, with results available on the JASPAR website. Additionally, the database update relied on complex software components, which were extensively revised and updated for each release.

The second objective was achieved by developing a pipeline called COBIND to discover TF co-binding patterns. The discovery method is based on non-negative matrix factorization (NMF) and automatically identifies potential TF co-binding patterns relative to known TFBSs. The pipeline was applied to 5,699 TFBS datasets for 401 TFs across seven species. The discovered co-binding patterns were shared among TFs from the same structural family and are likely functionally relevant, as they exhibit higher evolutionary conservation compared to isolated TFBSs. Furthermore, open chromatin data and single-molecule footprinting data supported the co-binding predictions in corresponding human cell lines and mouse embryonic stem cells, respectively.

Sammendrag

Transkripsjonsfaktorer (TF-er) spiller en avgjørende rolle i å regulere transkripsjonen av DNA til RNA. TF-er binder seg til regulatoriske DNA-regioner, og sikrer at spesifikke gener blir transkribert til rett tid og sted. Hovedmålene til TF-er er cis-regulatoriske regioner (CRR), kjent som promotere og enhancere. Promotorer er lokalisert rundt transkripsjonsstartsteder (TSS), mens forsterkere kan finnes i store lineære avstander fra genene de regulerer. TF-er binder promotere og forsterkere for å rekruttere andre proteiner og RNA-polymerase II for å starte transkripsjonen. TF-er er sekvensspesifikke; deres DNA-bindende domener (DBD) samhandler med korte DNA-nukleotidsekvenser kalt transkripsjonsfaktorbindings seter (TFBS). Å forstå genreguleringsmekanismer krever å undersøke hvordan TF-er samhandler med DNA og identifisere mønstrene de gjenkjenner.

Mens de bindende egenskapene til individuelle TF-er er godt dokumentert, er transkripsjonsregulering kompleks og involverer ofte flere TF-er som jobber sammen. Ulike kombinasjoner av TF-er kan samhandle med DNAet ved å binde samme region eller danne proteinkomplekser. Fordelingen av TFBS-er innenfor CRR-er bestemmer bindingen til forskjellige TF-er og kan ha en annen regulatorisk effekt. Orienteringen og avstanden mellom TFBS-er er kjent som DNA-motivgrammatikk. Mange cellulære prosesser, som celledifferensiering og respons på ytre stimuli, er avhengige av transkripsjonsregulering ved å samarbeide TF-er. Å studere disse kombinatoriske TF-bindingene er utfordrende, men avgjørende for å forstå den fulle kompleksiteten til transkripsjonsregulering.

Ekperimentelle teknikker som kromatinimmunutfelling etterfulgt av sekvensering (ChIP-seq) kan bestemme genomiske sekvenser bundet av en TF av interesse *in vivo*. Beregningsmetoder modellerer deretter TF-DNA-interaksjoner. Den mest brukte beregningsmodellen for TF-DNA-interaksjoner er posisjonsvektmatrisen (PWM). PWM-er oppsummerer eksperimentelt avledede sekvenser bundet av en TF og definerer et DNA-motiv gjenkjent av en bestemt TF. Motiver kan oppdages *de novo* fra et sett med genomiske sekvenser ved å identifisere overrepresenterte mønstre. Beregningsmodeller av TF-DNA-interaksjoner er lagret i databaser og kan brukes til å forutsi TFBS-er, kommentere ukjente motiver eller forutsi TF-sambindingshendelser. JASPAR er et eksempel på en database som

inneholder samlinger av TF-bindingsmotiver for TF-er fra forskjellige taksonomiske grupper.

Hovedmålet med denne oppgaven var å bruke beregningsmetoder for å utforske TF-DNA-interaksjoner ved (1) å oppdage nye TF-bindingsmotiver og oppdatere JASPAR-databasen, samtidig som man utvikler og oppdaterer robuste pipelines for automatisert databehandling og etterbehandling, og (2) identifisere og karakterisere co-bindende DNA-mønstre.

Det første målet ble oppnådd ved å bidra til de to siste utgivelsene av JASPAR-databasen. For begge utgivelsene ble det samlet inn en stor mengde offentlig tilgjengelige genomiske data for å oppdage DNA-bindende motiver. Disse motivene ble deretter manuelt kurert og lagt til CORE og UNVALIDATED-samlingene, avhengig av tilgjengelig ortogonal støtte. I tillegg til å legge til ny informasjon i databasen, ble eksisterende profiler revidert for å sikre metadatakvalitet. De endelige settene med motiver ble brukt til nedstrømsanalyser, med resultater tilgjengelig på JASPAR-nettstedet. I tillegg var databaseoppdateringen avhengig av komplekse programvarekomponenter, som ble omfattende revidert og oppdatert for hver utgivelse.

Det andre målet ble oppnådd ved å utvikle en rørledning kalt COBIND for å oppdage TF-sambindingsmønstre. Oppdagelsesmetoden er basert på ikke-negativ matrisefaktorisering (NMF) og identifiserer automatisk potensielle TF-sambindingsmønstre i forhold til kjente TFBS-er. Rørledningen ble brukt på 5 699 TFBS-datasett for 401 TF-er på tvers av syv arter. De oppdagede sambindingsmønstrene ble delt mellom TF-er fra samme strukturelle familie og er sannsynligvis funksjonelt relevante, da de viser høyere evolusjonær bevaring sammenlignet med isolerte TFBS-er. Videre støttet åpne kromatindata og enkeltmolekykfotavtryksdata sambindingsspådommene i henholdsvis tilsvarende humane cellerlinjer og museembryonale stamceller.

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Preface

This thesis is submitted in partial fulfillment of the requirements for the degree of *Philosophiae Doctor* at the University of Oslo. The presented work was conducted at the Centre for Molecular Medicine Norway (NCMM), organized by the University of Oslo, under the supervision of Professor Dr. Anthony Mathelier and the co-supervision of Associate Professor Dr. Jonas Paulsen. The Research Council of Norway funded this research through grants 187615 and 288404, the Norwegian Cancer Society through grant 197884, and the Nordic EMBL Partnership Hub for Molecular Medicine through NordForsk grant 96782.

A collection of three articles is the basis of this thesis. Two papers were published, and one is in peer review at the thesis submission date. The overarching theme of the presented articles is transcription factor binding to the DNA. Introductory chapters provide the necessary theoretical overview and motivation for this research.

Acknowledgements

Preface

Abbreviations

3'UTR	three prime untranslated region
5'UTR	five prime untranslated region
5mC	5-methylcytosine
A	adenine
ATAC-seq	assay for transposase-accessible chromatin using sequencing
bHLH	basic helix-loop-helix
bp	base pair
C	cytosine
CAP-SELEX	consecutive affinity-purification systematic evolution of ligands by exponential enrichment
CGI	CpG island
ChIP-seq	chromatin immunoprecipitation followed by sequencing
CNNs	convolutional neural networks
CRRs	cis-regulatory regions
DAP-seq	DNA affinity purification sequencing
DBD	DNA-binding domain
DHS	DNase I hypersensitivity
DNA	deoxyribonucleic acid
DNMTs	DNA methyltransferases
DOI	Digital Object Identifier
DWMs	dinucleotide weight matrices
EM	expectation maximization
eRNA	enhancer RNA

Preface

FAIR	Findable, Accessible, Interoperable, Reusable
G	guanine
GTFs	general transcription factors
H3K27ac	Histone 3 Lysine 27 acetylation
H3K27me3	Histone 3 Lysine 27 trimethylation
H3K36me	Histone 3 Lysine 36 methylation
H3K4me3	Histone 3 Lysine 4 trimethylation
H3K9me3	Histone 3 Lysine 9 trimethylation
H4K20me	Histone 4 Lysine 20 methylation
HMMs	hidden Markov models
IC	information content
kb	(or kbp) kilobase pair
lncRNAs	long non-coding RNAs
Mb	(or Mbp) megabase pair
mESCs	mouse embryonic stem cells
miRNAs	micro RNAs
mRNA	messenger RNA
ncRNAs	non-coding RNAs
NMF	non-negative matrix factorization
NTPs	nucleotide triphosphates
ORF	open reading frame
PBM	protein binding microarray
PDB	Protein Data Bank
PFM	position frequency matrix
PIC	preinitiation complex
piRNAs	PIWI-interacting RNAs
Pol II	RNA polymerase II
PPI	protein-protein interaction

PPM	position probability matrix
pre-mRNA	precursor mRNA
PSSMs	position specific scoring matrices
PTMs	post-translational modifications
PWMs	position weight matrices
RNA	ribonucleic acid
rRNA	ribosomal RNA
SELEX	systematic evolution of ligands by exponential enrichment
siRNAs	small-interfering RNAs
SMF	single-molecule footprinting
snoRNAs	small nucleolar RNAs
snRNAs	small nuclear RNAs
T	thymine
TADs	topologically associated domains
TAFs	TBP-associated factors
TBP	TATA-binding protein
TET	ten eleven translocation
TFBSs	transcription factor binding sites
TFFMs	transcription factor flexible models
TFs	transcription factors
tRNA	transfer RNA
TSSs	transcription start sites
U	uracil

Introduction

Chapter 1

General background

Multicellular organisms are composed of hundreds of cell types. Every cell in an organism has deoxyribonucleic acid (DNA) stored in the cell's nucleus. DNA is the hereditary material that contains the instructions for a cell to perform its functions, divide, and differentiate. The cells in an organism must also communicate with each other and the environment to coordinate all the processes. Four bases, adenine (A), thymine (T), cytosine (C), and guanine (G), are composing DNA sequences (genetic code) to encode a blueprint of life. All genetic material within a single cell is known as the genome. Specific regions of DNA, termed genes, are considered functional units of heredity. A subset of genes encode proteins, while other genes do not. Proteins are molecules that perform most of the cellular functions. A gene is first copied (transcribed) into ribonucleic acid (RNA) during transcription. In the case of protein-coding genes, RNA molecules are then used in translation to inform what amino acids should be brought and connected into a chain to form a protein. RNA molecules transcribed from non-coding genes are further processed and participate in other cellular processes, such as transcription regulation.

Every cell in an organism has the same set of genes. However, cells are different and form different tissues and organs. What enables cells to differentiate into many types when the DNA sequence is identical? Gene transcription is highly regulated, ensuring that specific cells express the right genes and gene combinations at the appropriate times and locations. Major players in this process are a particular class of proteins — transcription factors (TFs). They act by binding to specific DNA sequences (regulatory regions), which can control the associated gene transcription rate. Interactions between TFs and regulatory DNA sequences ensure gene transcription control in space and time. This chapter overviews basic molecular biology concepts, such as DNA, RNA, genome, and the process of transcription, to provide the background to dive deeper into how transcription is regulated.

1.1 Genome organization

In eukaryotes, the nucleus stores hereditary information as DNA. Cells copy (replicate) genomic DNA during division or transcribe portions of the DNA (genes) into RNA to produce proteins. Human DNA is only 2 nanometers wide, but if

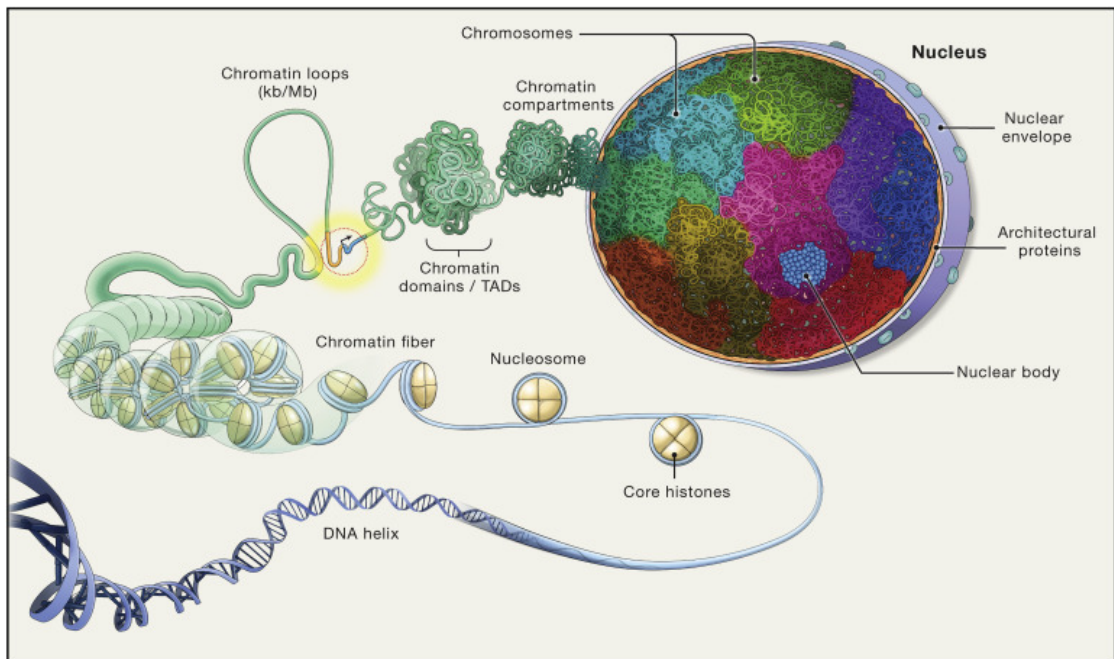


Figure 1.1: Eukaryotic genome organization. Genomic DNA is wrapped around histones to form nucleosomes, the basic units of chromatin fiber. Chromatin can form loops that span kilobase pairs (kb) or megabase pairs (Mb) and are essential for gene expression regulation. Bigger stretches of chromatin are organized into chromatin domains or topologically associated domains (TADs), forming chromatin compartments to pack everything into the nucleus. Each chromosome occupies a territory in the nucleus. The nucleus also contains a nuclear body and architectural proteins. A nuclear envelope surrounds the contents of the nucleus. Adapted from [2].

we stretch it into a single thread, it would be around 2 meters long [1]. The microscopic cell nucleus stores tightly compacted DNA. Multiple levels of genome organization exist. The simplest level of organization is a DNA helix wrapped around a specific class of proteins called histones. The highest level of organization is a chromosome, visible during the cell division. A chromosome consists of a single molecule of DNA and various proteins. The human genome is organized into 22 pairs of autosomal chromosomes and one pair of sex chromosomes (XX or XY). The DNA organizes into chromosome territories in the nucleus (Figure 1.1) [2]. In addition to the DNA and architectural proteins, the nucleus contains different types of nuclear bodies (nucleolus, paraspeckles), made of proteins and involved in cellular processes, such as telomere maintenance or stress response [2, 3]. A nuclear envelope surrounds the content of the nucleus. The specific organization ensures proper genome function — expressing genes when and where needed.

1.1.1 DNA

The existence of DNA was discovered in 1869 by Friedrich Miescher when he found a new substance inside the nuclei of the human white blood cells that did not have the same properties as proteins or lipids [4, 5]. In 1953, contributions of Rosalind Franklin, James Watson, and Francis Crick led to the publication of resolved DNA

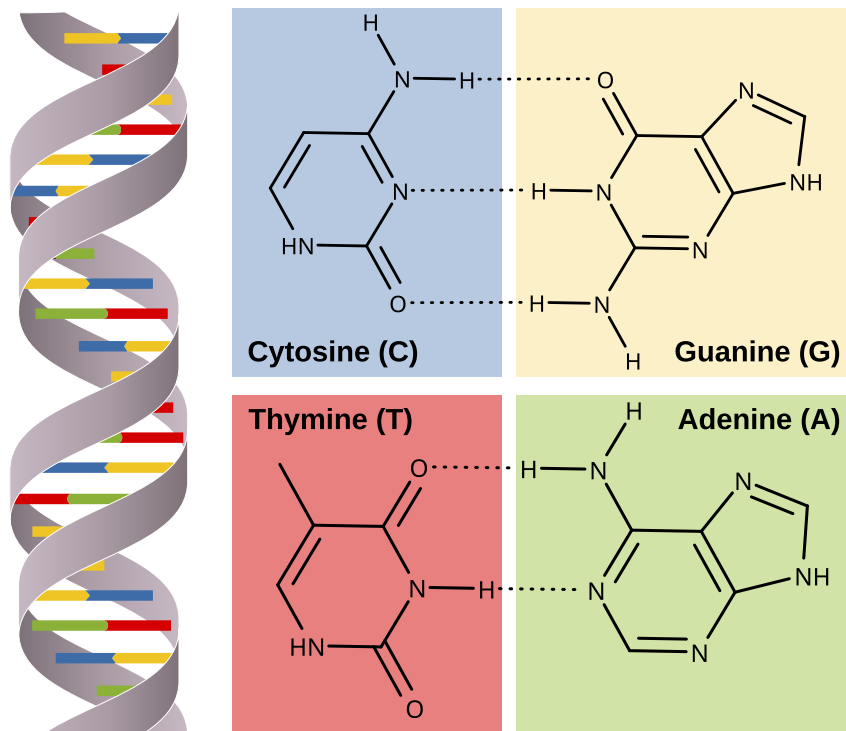


Figure 1.2: DNA structure. The two chains of the DNA are composed of a sugar backbone and four nucleotides: adenine (A), thymine (T), cytosine (C), and guanine (G). Nucleotides in the two chains connect via hydrogen bonds in a strict complementarity, where A is bound to T and C is bound to G.

molecule structure and is often considered the beginning of molecular biology [6]. This fundamental moment in science history allowed us to explain the possible mechanism of DNA replication, thus information multiplication and transfer.

The DNA stores genetic information in the cell. In the DNA structure, the sugar backbone (deoxyribose) holds four main types of nucleotides: purines adenine (A) and guanine (G), and pyrimidines thymine (T) and cytosine (C) [7] [1]. Chargaff's rule states that in the DNA of any species, the proportions of A and T nucleotides and C and G are always equal, demonstrating unique base-pairing (complementarity), when forming a double-stranded DNA helix [7, 8]. A-T and C-G nucleotide pairs in two strands of the DNA and are held by two and three hydrogen bonds, respectively (Figure 1.2).

The most commonly found DNA structure is called a B form of DNA. Two helical DNA strands entwine to form spaces between the strands, major and minor grooves, which are 22 and 12 Å wide, respectively. The grooves create space for the proteins to interact with the double-stranded DNA. One of the most important classes of proteins that bind DNA is transcription factors (TFs) [9]. TFs are essential for the regulation of transcription. They act by recognizing specific sequences of nucleotides (see next chapter, where TF-DNA interactions are described in detail).

1.1.2 Chromatin

The chromatin is a mixture of DNA and proteins that form chromosomes, the highest level of genome organization. The fundamental element of chromatin is a nucleosome, which consists of 146 bp of DNA wrapped around the core histone protein octamer. Nucleosomes gather into a chromatin fiber typically 5-24 nanometers in diameter. The fibers organize into higher structures. They fold into chromatin domains, called topologically associated domains (TADs) [10]. TADs associate with each other and create chromatin compartments. When cells are dividing, chromatin assembles into a metaphase chromosome. The chromatin is organized into domains and compartments and tends to localize in specific territories (Figure 1.1) [2, 11]. Chromatin fibers from different chromosomes interact at the edges but do not entangle. In addition to chromatin, the nucleus contains protein structures called nuclear bodies. Nuclear bodies, such as nucleolus, Cajal bodies, paraspeckles, and others, are membrane-less organelles [12]. Their functions include the coordination and processing of RNAs and coordination of chromatin-modifying proteins, among others [12]. The nuclear envelope surrounds the contents of the nucleus.

The canonical core histone proteins are H2A, H2B, H3, and H4, with two copies each in a single octameric histone core (Figure 1.1). A linker histone H1 separates every core nucleosome [11, 13]. Variants in specific genomic regions replace well-conserved canonical histone proteins. For example, H3.3 replaces the H3 histone in active genes and promoter regions [13, 14]. Histone protein N-terminal tails undergo extensive post-translational modifications (PTMs), such as acetylation and methylation, on their amino acids. These modifications are essential for controlling the accessibility of the chromatin. Accessible chromatin is accessed by a transcriptional machinery and regulatory proteins [13].

Based on its activity state, the chromatin can be heterochromatin or euchromatin [15]. Euchromatin is a more open, active chromatin with more widely spaced nucleosomes. Several histone tail chemical modifications (Histone 3 Lysine 4 trimethylation (H3K4me3), Histone 3 Lysine 27 acetylation (H3K27ac), Histone 3 Lysine 36 methylation (H3K36me)) and histone variants (H3.3, H2A.Z, H2A.B) are marks of active transcription. Heterochromatin is generally known as a silent chromatin and is less accessible and transcribed. Constitutive heterochromatin is more static and contains Histone 3 Lysine 9 trimethylation (H3K9me3). Histone 3 Lysine 27 trimethylation (H3K27me3) marks the facultative heterochromatin, which can switch between active and inactive states. For example, facultative heterochromatin marks an inactive X-chromosome [11].

1.2 Transcription

Genes represent basic units of heredity. A gene is a stretch of DNA in the genome subjected to transcription by RNA polymerase. Transcription is a process of copying DNA into an RNA molecule. Genes can be protein-coding or non-coding. The protein-coding RNA template is translated into a protein, while non-coding

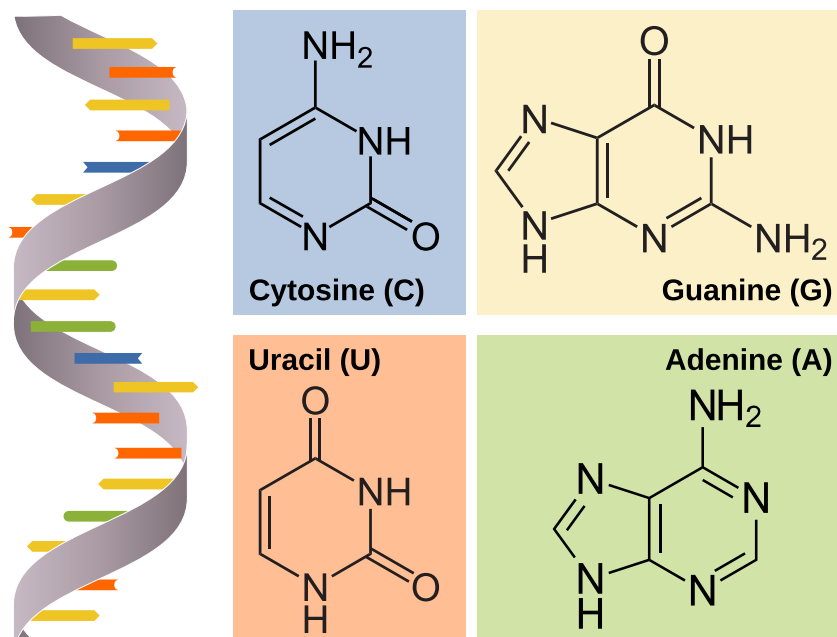


Figure 1.3: RNA structure. RNA is a single-stranded molecule composed of a sugar backbone and four nucleotides: uracil (U), adenine (A), cytosine (C), and guanine(G).

genes produce non-coding RNA molecules. A collection of proteins assembles into transcriptional machinery and organizes the transcription process.

1.2.1 RNA

The transcription machinery, with the leading player RNA polymerase, transcribes a portion of the genome to produce RNA. An RNA molecule is single-stranded and consists of a sugar backbone (ribonucleic acid) and four types of nucleotides: uracil (U), A, C, and G (Figure 1.3). RNA molecules can adopt various secondary structures by forming hydrogen bonds between nucleotides (C pairs with G, while A pairs with U) within the same molecule [16].

The main types of RNAs, based on their functions, are messenger RNA (mRNA), ribosomal RNA (rRNA), transfer RNA (tRNA), and other non-coding RNAs (ncRNAs) [16–18]. A precursor mRNA (pre-mRNA) is produced in the nucleus during transcription by copying a protein-coding gene (Figure 1.4) [17]. RNA polymerase synthesizes RNA from nucleotide triphosphates (NTPs) (ATP, GTP, CTP, and UTP, precursors of A, G, C, and U, respectively). A pre-mRNA is heavily processed to produce an mRNA that carries information to the cytoplasm, where ribosomes translate it to a polypeptide. A mRNA has an open reading frame (ORF), consisting of introns and exons. The ORF is flanked by three prime untranslated region (3'UTR) and five prime untranslated region (5'UTR) [16, 19]. During splicing, a specific protein complex, the spliceosome, processes most of the pre-mRNA by cutting out intronic (non-protein coding) gene segments to leave exons [20]. Introns can be spliced differently by choosing to leave out some exons, creating a variety of transcripts copied from the same gene. This process

is known as alternative splicing [21]. Splicing factor proteins recognize splicing signaling sequences at the intron-exon boundaries [20, 21]. In addition to splicing, a cap is added at the 5' end of the RNA, while multiple added A nucleotides at the 3' end form a poly-A tail [21]. After this, mRNA travels from the nucleus to the ribosomes in the cytoplasm. Ribosomes receive individual mRNAs and read the information encoded. Ribosomes contain rRNA, and each amino acid is brought into the ribosome by tRNAs that recognize every three successive mRNA nucleotides, termed codons [16, 18]. Codons code for amino acids.

Longer than 200 nucleotides non-coding RNAs (long non-coding RNAs (lncRNAs)) and shorter small ncRNAs are the two main size-based categories of ncRNAs. Small ncRNAs can be divided into micro RNAs (miRNAs), small nucleolar RNAs (snoRNAs), small nuclear RNAs (snRNAs), small-interfering RNAs (siRNAs) and PIWI-interacting RNAs (piRNAs) [22]. miRNAs have an essential function in eukaryotic gene regulation. They can target and silence mRNAs to prevent translating them into proteins [17]. Aberrations of the ncRNA activities can be associated with diseases such as cancer. For example, miRNA can act as a tumor suppressor and target genes responsible for cell growth. snRNAs and snoRNAs are involved in splicing. snoRNAs can also guide the modifications of rRNAs [22]. The primary function of siRNAs is silencing gene expression post-transcriptionally. piRNAs play a role in silencing transposable elements [23].

1.2.2 RNA polymerase and transcriptional machinery

In eukaryotes, RNA polymerase II (Pol II) is responsible for transcribing genes encoding mRNAs and some ncRNAs [24]. This twelve-subunit protein uses NTPs as a substrate to produce RNA molecules from a DNA template. The RNA sequence synthesis process follows nucleotide complementarity.

Pol II is not able to recognize the transcription start sites (TSSs) by itself and needs to be guided by a set of proteins known as general transcription factors (GTFs) (Figure 1.4). Pol II and GTFs form a minimal transcription machinery, and all these proteins, together with the template DNA, form the preinitiation complex (PIC) [24, 25]. The GTFs are TFIIB, TFIID, TFIIE, TFIIF, and TFIIH, and each plays a specific role in transcription initiation. They assemble one after another. Proteins TATA-binding protein (TBP) and TBP-associated factors (TAFs) compose TFIID and recognize a particular sequence of DNA near some TSSs, called a TATA box [26]. TATA box, however, is present at only 20% of eukaryotic promoters. In other cases, TBP binds TATA-like sequences that contain 1 or 2 mismatches from the consensus [26]. TFIIB and the non-phosphorylated form of Pol II join the complex, followed by TFIIF, TFIIE, and TFIIH. Finally, Pol II is phosphorylated to proceed to elongation [24]. The transcriptional machinery produces a central portion of RNA during transcription elongation. Transcription ends with the transcription termination process, and RNA travels for further processing and splicing.

Different cell types express a limited set of genes called housekeeping genes. However, for cells to differentiate and perform cell-type specific functions, specific genes need to be transcribed at a particular time and place. The basic

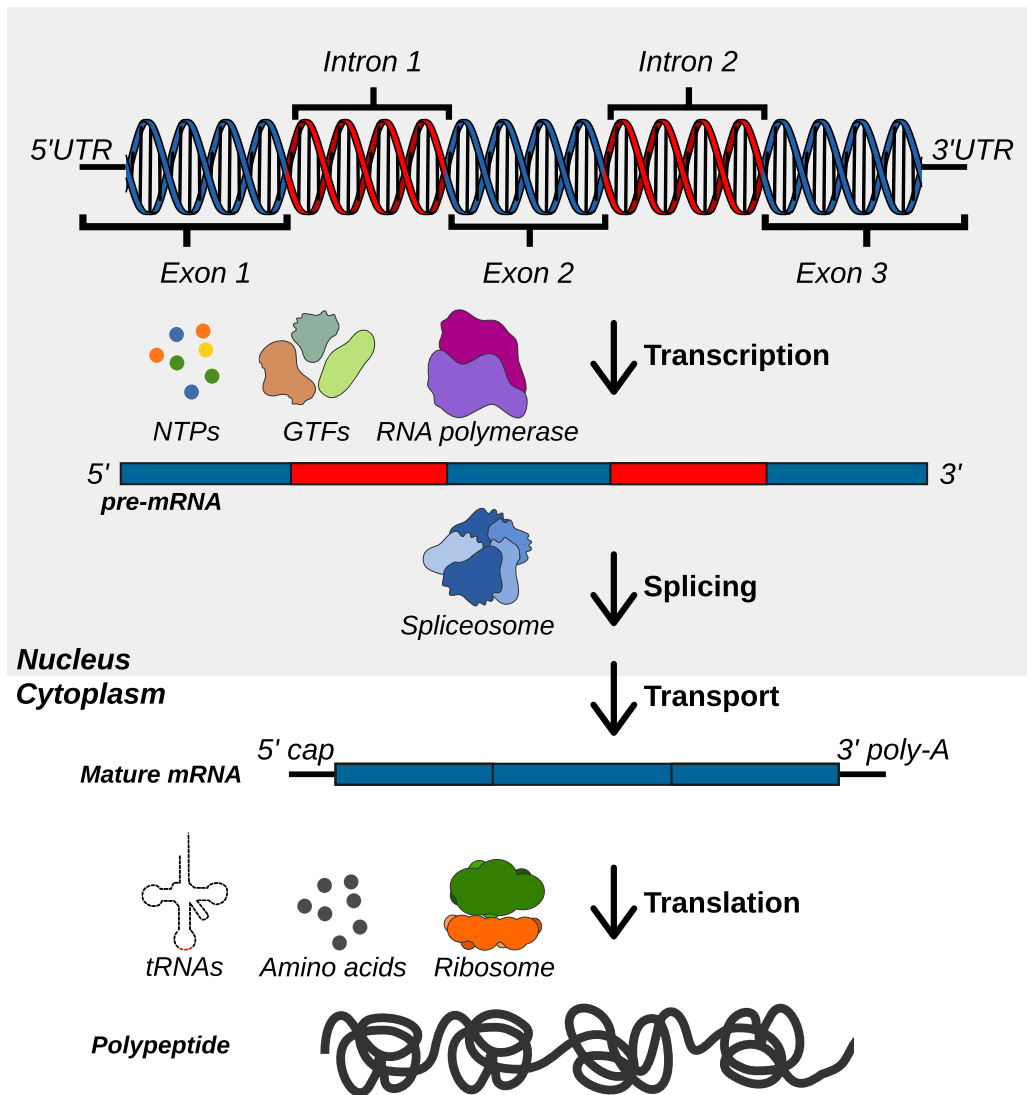


Figure 1.4: From gene to protein. During transcription, an open reading frame (ORF), flanked by 5' and 3' untranslated regions (UTRs), is transcribed by RNA polymerase to produce pre-mRNA. General transcription factors (GTFs) are part of the transcription complex, and RNA polymerase uses nucleotide triphosphates to produce the RNA molecule. Pre-mRNA has introns and exons. The spliceosome cuts out non-protein coding introns during splicing, leaving only protein-coding exons. 5' cap and 3' poly-A tail are added to produce a mature mRNA. In the cytoplasm, the ribosome translates mRNA into a polypeptide chain. tRNAs bring each amino acid into the ribosomes.

Chapter 1. General background

transcriptional machinery cannot decide what genes to express, but various regulatory processes make those instructions. For example, TFs bind DNA at regulatory regions. These regions are promoters (proximal to the TSS), enhancers (distal from the TSS), silencers, and insulators. TF-DNA interactions signal the PIC assembly at the associated gene promoter. Mechanisms of TF binding to the DNA and transcriptional regulation are very complex, and the field still lacks details on the organization of these processes. The next chapter overviews the regulatory regions, TFs, and transcription regulation.

Chapter 3

Computational modeling of transcription factor-DNA interactions

Researchers employ various experimental methods to study how transcription factors interact with DNA, identifying sequences or genomic regions that indicate TF specificity for particular DNA patterns. This data serves as the foundation for the computational modeling of TF-DNA interactions. Different models are available today, from simple and widely used position weight matrices (PWMs) to advanced deep learning models. This chapter will cover different models and techniques used to create them, highlighting resources that store these models and explore their applications in research.

3.1 Transcription factor DNA binding models

3.1.1 Position weight matrices (PWMs)

The most popular and widely used representation of TF-DNA interactions is the PWMs (also called weight matrices or position specific scoring matrices (PSSMs)) (Figure 3.13) [68, 96, 97]. Several steps are followed when building a PWM.

(1) From a set of aligned sequences (for example, processed experimental data or sequences collected from literature) that are bound by a TF of interest, a position frequency matrix (PFM) is computed (Figure 3.13A-B) [98]. A PFM, or a count matrix, is created by counting the occurrences of each nucleotide at each position [27, 96, 99]. The length of this matrix corresponds to the DNA motif length recognized by the TF [68]. This matrix quantitatively describes the known TFBSs [27].

(2) Counts can be converted into probabilities to build a position probability matrix (PPM) (Figure 3.13C) using

$$p(b, i) = \frac{f_{b,i} + s(b)}{N + \sum_{b' \in \{A,C,G,T\}} s(b')} \quad (3.1)$$

where $p(b, i)$ is a probability of base b in position i ; $f_{b,i}$ is counts of base b in position i ; $s(b)$ is a pseudocount function; N is the number of sites. Pseudocounts

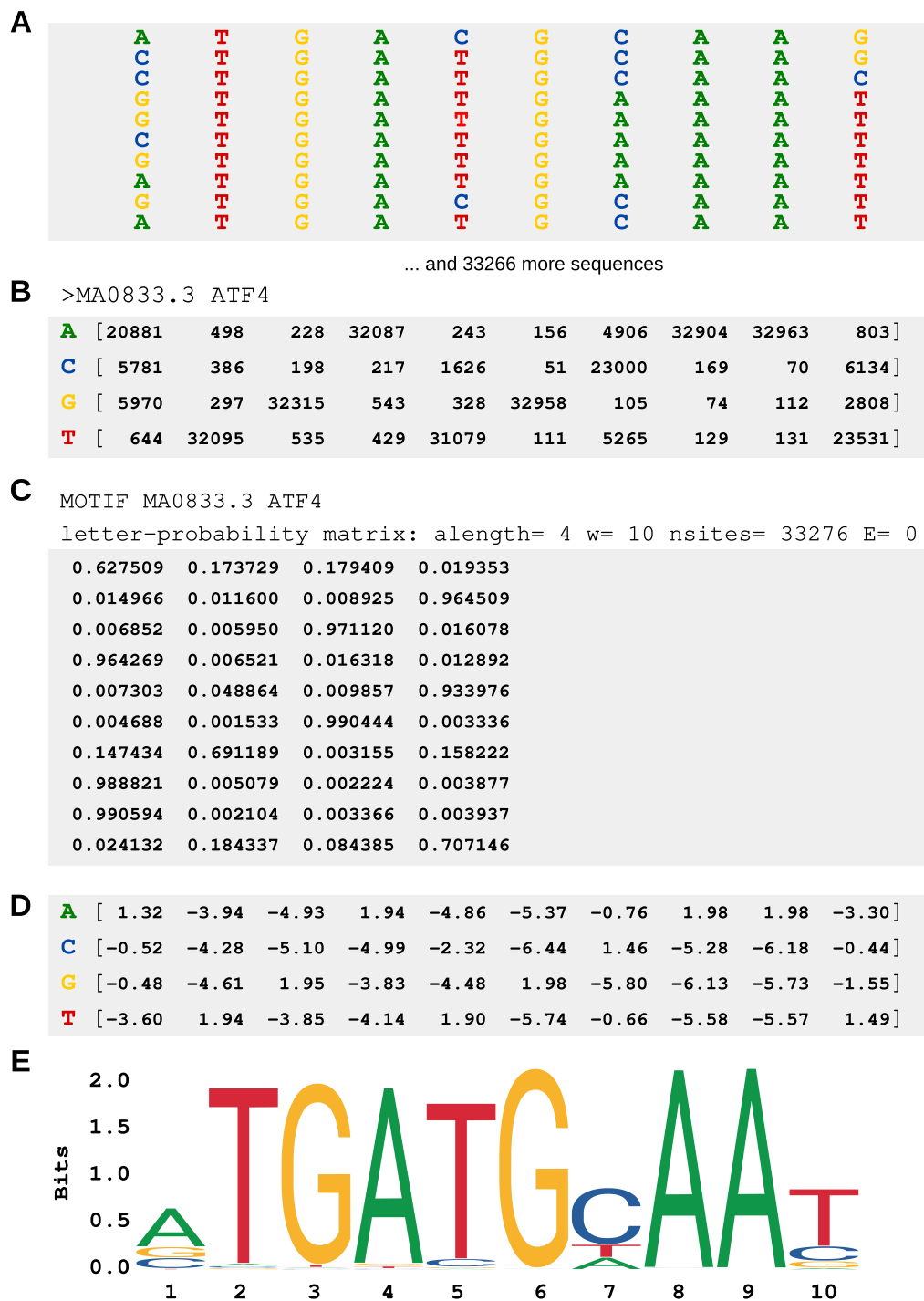


Figure 3.13: Modeling the binding of ATF4 transcription factor. **A.** A set of aligned sequences bound by ATF4. Various experimental techniques can produce sequences. **B.** Position frequency matrix (PFM), which represents the nucleotide occurrence counts in each sequence position (visualized in JASPAR format). **C.** PFM can be converted into a position probability matrix (PPM) (visualized in MEME format). **D.** Position weight matrix (PWM) computed from the PPM. **E.** A sequence logo visualizes information content (IC) for the ATF4 binding profile. High values indicate a strong preference toward a particular base. Matrices from the JASPAR database ID MA0833.3.

can be added to each cell of the PFM to eliminate zeros [27].

(3) A PWM (Figure 3.13D) is constructed by dividing the nucleotide probabilities by expected background probabilities and converting the resulting values to a log scale (known as log odds ratio). The most straightforward model assumed equal 0.25 background probabilities for each base. The computation is done using

$$W_{b,i} = \log_2 \frac{p(b,i)}{p(b)} \quad (3.2)$$

where $p(b)$ is the background probability of base b ; $p(b,i)$ is the probability of base b in position i ; $W_{b,i}$ is the PWM value for base b in position i .

Frequency or count matrices can be represented in multiple formats. For example, JASPAR format (Figure 3.13B) visualizes the counts, while MEME format visualizes the probabilities of each nucleotide in each position (Figure 3.13C). Some computational tools require specific format matrices, so familiarity with different count and probability representations is needed.

It is difficult to interpret the actual DNA pattern preferentially bound by a TF only by looking at the PWM or PFM matrices. Therefore, sequence logos are used for visualization (Figure 3.13E). The sequence logo gives information about the frequency of each base at each position and the deviation from the expectation. Sequence logo visualizes information content (IC) in bits. IC is a basic unit of information and quantifies how much information a binding site provides about the TF binding preferences. IC can be computed from probabilities using

$$D_i = 2 + \sum_b p_{b,i} \log_2 p_{b,i} \quad (3.3)$$

where D_i is the IC at position i , $p(b,i)$ is the probability of base b in position i .

All computational models used to model TF-DNA interactions are based on certain assumptions. One of the most significant limitations of PWMs is that these models assume the independence of nucleotides at different pairings [27, 96, 100]. Locations of nucleotides at adjoining positions can be important for transcription factor binding, as was demonstrated through biochemical, statistical, and crystal structure analysis [100]. Another potential caveat of the PWMs is their inability to accurately model compositional binding, as homodimers and heterodimers can have varying distances between the two binding sites [27, 100]. The limitations of PWMs call for more complex models. One can expand the PWM model to capture additional information, such as DNA methylation [101]. Other TF-DNA interaction models are described hereafter. However, despite their limitations, the availability and ease of use keep PFMs and PWMs the primary methods to model TF-DNA interactions.

3.1.2 Other more complex models

Other more complex models were introduced to address the limitations of PWMs. For example, the dinucleotide weight matrices (DWMs) extend the PWMs by considering pairs of adjacent nucleotides (dinucleotides) instead of individual

nucleotides [102]. DMWs were shown to be accurate representations of TF binding sites, performing well when predicting those TFBSs. However, a DMW will contain the same information as the PWM if positions are completely uncorrelated [102].

In 2013, hidden Markov models (HMMs) based transcription factor flexible models (TFFMs) were introduced. They model both successive position interdependence within TFBSs and variable length motifs, which simple PWM models cannot achieve [100]. It was demonstrated through protein binding array studies that dependencies are more substantial between the neighboring positions in the binding site compared to others [103]. In first-order TFFMs, an HMM state represents each position within a TFBS. Each HMM state in the first order-TFFM is then decomposed into four states (one per nucleotide) to build a detailed TFFM [100]. TFFMs are built using experimentally validated TF-DNA interaction sequences.

Machine learning, deep learning, in particular, is the most state-of-the-art approach for TF-DNA interaction modeling. These approaches address critical limitations of the simpler TF-DNA binding models [104]. In a deep-learning framework predicting activity from a sequence, one-hot encoded DNA sequences are used as input (Figure 3.14A) [105]. One needs to prepare training, validation (used for selecting methods, hyperparameter tuning, and avoiding overfitting), and testing datasets (used for model evaluation) [106]. Different types of neural network architectures are built to train the deep learning models [106]. For example, convolutional neural networks (CNNs) have convolutional layers to detect the motifs, pooling layers to reduce the dimensionality, and fully connected layers to make predictions about TF binding affinity (Figure 3.14B) [105]. The models are trained to classify binding vs. non-binding or predict the strength of the binding (Figure 3.14C). Finally, the model's predictive performance is evaluated. To describe the motifs and sequence features, the model needs to be interpreted by understanding the inner workings of the so-called "black box" models (Figure 3.14D) [105]. For instance, one can derive PWMs from the first-layer neurons encoded in convolutional weight matrices. A more common strategy is to search for subsequences that activate a layer and then construct a PWM from aligned activating sequences [105]. Interpretation of the models often requires additional tools based on certain assumptions, limiting deep learning-based approaches [105].

The pioneering approach created to detect motifs and predict TF binding sites is DeepBind [107]. Another method, DeepSea, was developed to predict multiple chromatin features in addition to the TF binding events [106, 108]. The deep learning approaches are particularly relevant for deciphering and modeling cooperative TF binding. One of the latest approaches used to predict binding profiles of TFs and demonstrated cooperative action is BPNet [72]. BPNet is a CNN and models the TF binding at base resolution. It consists of multiple convolutional layers and predicts the TF binding signal in mouse embryonic stem cells (mESCs) for four TFs (Oct4, Sox2, Nanog, and Klf4). To interpret the results from the BPNet and retrieve the DNA motifs, another tool, TF-MoDISco, is required [109]. The recurrent patterns are highlighted by aggregating the importance scores across aligned high-scoring regions in the sequences. Candidate consensus motifs are identified by clustering and aligning

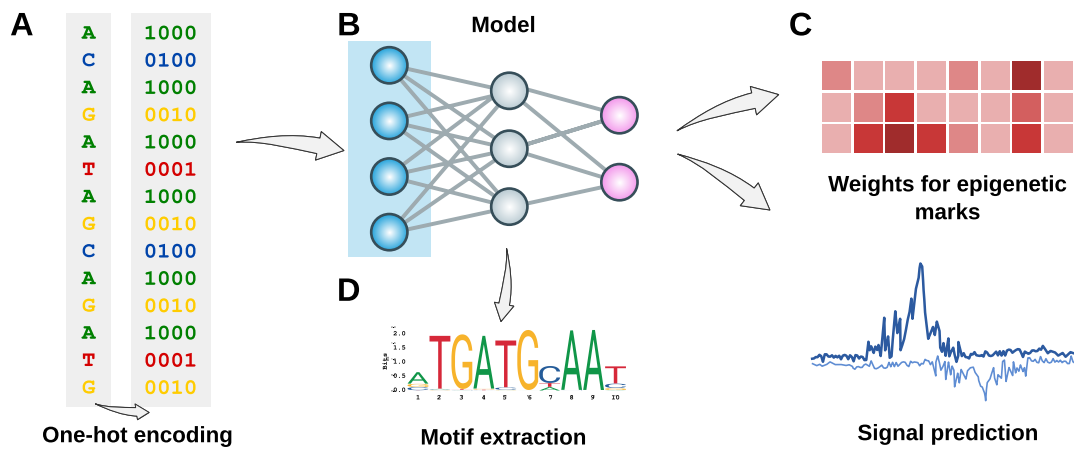


Figure 3.14: Deep learning approach to model TF-DNA interactions. **A.** DNA sequences are one-hot encoded before being used as input to the model. **B.** The neural network consists of multiple layers that make predictions. **C.** The model can predict the scores for epigenetic marks and the signal of TF binding. **D.** DNA binding motifs can be extracted from the network layers by interpreting the deep learning models.

high-scoring sequence segments. Motifs are then filtered and scored based on their importance and prevalence. Finally, motifs can be visualized for more straightforward interpretation and annotated using collections of known motifs. In their study, the authors of BPNet demonstrated that their analyzed TFs are binding using soft syntax rules, which means there are no strict requirements on the numbers of nucleotides between the two TFs co-binding the region. The results also showed that Nanog binds with helical periodicity and TF cooperativity directionally [72].

3.1.3 *De novo* motif discovery

When analyzing TF-DNA interaction data, such as ChIP-seq, one can find over-represented sequence motifs using existing collections of DNA binding motifs (see the section on databases of TF-DNA binding models). However, motif collections are always limited, so to discover new motifs, *de novo* motif discovery approaches have been developed [66]. *De novo* motif discovery is a process inferring enriched DNA patterns in a set of DNA sequences compared to the background expectations [99]. The aim of *de novo* motif discovery is to find short, statistically over-represented DNA patterns in the input set of sequences [110, 111]. The used background can be a Markov model with nucleotide frequencies similar to the input or random sequences, such as shuffled original input sequences. Algorithms discover motifs in sequences derived from experimental methods, such as SELEX, PBM, and ChIP-based techniques [66].

"Word-based" approaches gather approximate occurrences of a k -mer in the input sequences and rank them based on their over-representation. RSAT *peak-motifs* relies on several of such algorithms, namely oligo-, position-, and dyad-analysis (Table 3.1) [112–114]. Oligo-analysis detects over-

represented oligonucleotides, while position-analysis identifies oligonucleotides with heterogeneous positional distributions in a provided set of sequences [112, 115, 116]. By default, word-based analysis performs 6 and 7 bp length motif search [113]. Dyad-analysis detects spaced motifs, which dimeric TFs can bind [112, 117]. Researchers observed that regulatory sites in the yeast genome often contain pairs of highly conserved trinucleotides, separated by non-conserved positions [117]. RSAT *peak-motifs* can also perform differential analysis (oligo-diff) when the control set is provided [113].

On the other hand, "profile-based" algorithms iteratively optimize an initial PWM and perform heuristic searches. These methods select positions from the input set, align the sequences, and then build and score the PWM [110]. For example, the computational tool MEME starts the motif search with multiple k -mers, then it selects the current best to be optimized using expectation maximization (EM) [118, 119]. One of the main limitations is that the presence of the motif is assumed in each input sequence [110]. STREME is the newest motif discovery tool from the authors of MEME that allows larger sets of input sequences (Table 3.1) [120].

Table 3.1: An overview of *de novo* motif discovery tools. Several computational tools are developed to find overrepresented sequence patterns.

Tool	Description	Reference
MEME	Profile-based algorithm; uses expectation maximization (EM); finds recurring, fixed-length, ungapped patterns.	[118]
STREME	Profile-based algorithm; uses suffix tree-based optimization; finds recurring, fixed-length (3-30 bp), ungapped patterns.	[120]
RSAT	Word-based algorithm; oligo-, position-, and dyad-analysis; finds variable size motifs.	[112–114]

All the above-mentioned computational tools have their strengths and weaknesses. For example, RSAT integrates a useful additional feature that compares discovered motifs with known motifs (read more about motif comparison in the section about the use of TF-DNA binding models). However, one of the main limitations is that these tools (except the RSAT dyad-analysis algorithm) aim to discover a single over-represented pattern in a set of sequences at a time. Specialized approaches would be needed to examine the relationship between different discovered patterns. The co-occurring patterns can indicate the TF cooperative binding.

3.1.4 Non-negative matrix factorization for motif discovery

A different approach from the above, non-negative matrix factorization (NMF), could be used for DNA pattern discovery in sequences. Matrix factorization (also referred to as matrix decomposition) is a class of unsupervised techniques that

infers low-dimensional structure from high-dimensional data [121]. NMF is one of the matrix factorization techniques. All matrix factorization algorithms generally learn two matrices: pattern and amplitude [121].

Before applying NMF on genomic sequences, the usual practice is to one-hot encode the nucleotide sequences. It is a classical categorical data processing step for machine learning approaches. Each nucleotide is assigned a unique four-bit long code (Figure 3.15A). Individual sequences are converted into a numerical array, which then can be passed to the NMF algorithm (Figure 3.15B). A number of patterns that the NMF needs to learn is a hyperparameter and needs to be pre-defined. The learned pattern matrix represents the relationships between the nucleotides [121]. Each row in the pattern matrix contains the contribution weight of each nucleotide at each position in the pattern. The amplitude matrix describes relationships between the sequences [121]. Each column in the amplitude matrix stores continuous weights that describe the relative contribution of each sequence in each inferred pattern. Each pattern gets a set of sequences assigned with the highest contribution scores. The two matrices can be used to distinguish subsets of sequences with particular patterns that can be represented as DNA sequence motifs. NMF has a hyperparameter k , instructing how many patterns should be learned. For example, with $k = 4$, four DNA patterns are learned from an input set of DNA sequences (Figure 3.15C).

NMF is the basis of a computational tool, seqArchR, to identify position-specific DNA patterns in promoter regions and decipher promoter architectures [122]. It is known that specific DNA motifs inform the selection of TSSs and are located at fixed distances from them. The seqArchR package can identify TSS-directing and lineage-specific motifs, such as TATA or nucleosome position signals. Defining combinations of these motifs can be used to classify promoter architectures and associate them with specific gene regulatory programs [122].

A limitation of NMF for motif discovery is that the algorithm takes arrays as the input, meaning the input genomic sequences must be the same length. Another significant limitation is that sequence patterns that NMF can distinguish have to occur at fixed distances. Various patterns can be discovered in separate patterns, but there must be enough sequences with the same pattern at the exact location. Lastly, a hyperparameter k can be challenging to define. While several methods have been introduced to estimate the cophenetic correlation coefficient to determine the number of patterns, there is no convergence on an optimal method [123].

3.2 Databases of TF-DNA interaction models

Several repositories store PWMs. For example, JASPAR, TRANSFAC, HOCOMOCO, and CIS-BP (Table 3.2) [124–128]. For the users of such resources, it is essential to understand what data is stored and how it is retrieved and processed. JASPAR is an open-access database and stores DNA binding motifs for TFs from six taxa. All motifs in JASPAR are manually curated. JASPAR discovers motifs from experimental data, such as ChIP-seq or DAP-seq, or mined as PFMs from various resources (for instance, a selection of CIS-BP database

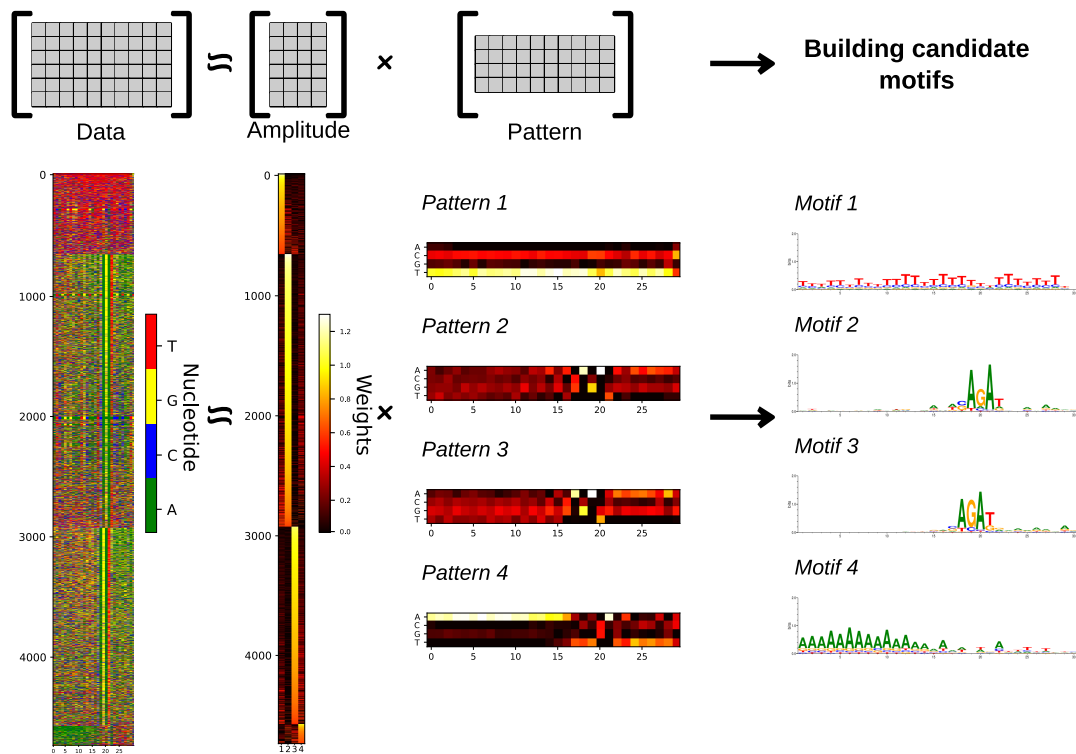


Figure 3.15: Application of non-negative matrix factorization for DNA pattern discovery in sequences. An example of 4 patterns NMF applied to a set of one-hot encoded DNA sequences (data matrix). The data matrix is factorized into amplitude and pattern matrices. Candidate DNA patterns (motifs) can be built using the resulting matrices. Each pattern produces one motif. Adapted from [95].

motifs) and publications. Independent evidence found by curators supports CORE collection motifs. In addition to motif collections, JASPAR provides analyzed data, such as motif clusters and various tools [124]. TRANSFAC also stores curated DNA motifs with experimentally determined and predicted binding sites. However, the complete database is not open-access [125, 126]. HOCOMOCO is a publicly available resource that compiles and stores human and mouse TF binding motifs. The motifs in the latest release of the database are human-curated and automatically benchmarked [127]. Finally, CIS-BP is also open-access and contains motifs for more than a thousand TFs from 131 diverse eukaryotes. Motifs are generated using PBM, SELEX, or CHIP-seq data but are also mined from other motif resources, such as TRANSFAC and JASPAR. A unique feature of CIS-BP is that the database offers inferred motifs. Based on the similarity of the DBDs, two TFs are binding similar motifs. Therefore, it is possible to assume a binding profile from one TF to a closely related other TF [128].

Table 3.2: An overview of DNA binding motif databases.

Database	Content	Reference
JASPAR	Open-access manually curated collections of DNA binding motifs for six taxonomic groups. The CORE collection contains motifs with orthogonal support.	[124]
TRANSFAC	Curated collection of TF binding motifs.	[126]
HOCOMOCO	Open-access manually curated motif collections of human and mouse TFs.	[127]
CIS-BP	Open-access database with a comprehensive list of predicted TFs from >300 species. Motifs are available if known.	[48]

Even if current databases contain thousands of motifs from various species, it is essential to keep these resources updated by adding new data and refining the old. Finally, new ways of TF-DNA modeling (such as deep-learning models) motivate the expansion or creation of new databases that could offer a variety of TF-DNA interaction models to the community.

3.3 Use of position weight matrices

Researchers use the collections of known TF binding profiles in numerous ways. One of the most critical applications is the prediction of TF binding sites. Computational tools predict putative TFBSs in the entire genome or perform TF binding enrichment in a set of regions. Another popular application of the TF motifs is an annotation of *de novo* discovered motifs, which is needed when no binding preferences are known for a TF of interest.

3.3.1 Prediction of TF binding sites

TF-DNA binding models can be used to predict putative transcription factor binding sites in the entire genome (also referred to as motif scanning) or a set of regions of interest (for example, ChIP-seq peaks or promoters) [66]. The TF binding models for predicting can be taken from various databases overviewed in the previous section [124–126].

In TFBS prediction, algorithms first score sequences using a PWM. The sliding window approach is used to scan a DNA sequence of interest. Each window (same size as a PWM) is scored. When scoring, the weights of each corresponding nucleotide in a PWM are summed to estimate the TF binding probability in a given DNA sequence using

$$S = \sum_{i=1}^w W_{l_i,i} \quad (3.4)$$

where S is a score for input sequence; w is the width of the PWM; l_i is the position i in the input sequence [27]. The final set of TFBS predictions is separated from the background and defined after applying the score threshold. Usually, statistical significance (measured through p -value) is evaluated to solve the thresholding issue. For example, it can be the computation of the probability to observe at least the same number of motif instances with a specific cutoff in a random sequence with total length (equal to the total size of sequences in the set of regions of interest) [66]. This task becomes difficult computationally if one tries to score multiple motifs (possibly with overlaps), which is relevant when investigating TF cooperativity and co-binding.

The prediction tool will find fewer motif instances with the higher score cutoff, and the resulting binding sites will likely correspond to the strong binding affinity. Compared with the lower cutoff, the algorithm predicts more sites, and they might represent both strong and weak binding sites [66]. However, the "futility theorem" states that predicted TFBSs outnumber the sites that are actually occupied by a TF [27]. False positively predicted TFBSs are not available for binding due to chromatin structure or local epigenetic properties [100].

Resources, such as JASPAR, in addition to the collections of PFMs, offer TFBS predictions for the most popular genome assemblies [124]. Bioinformaticians designed various tools that scan or find over-represented motifs in regions. For example, PWMScan, FIMO, or HOMER [47, 129, 130].

3.3.2 Prediction of direct TF binding sites

According to the "futility theorem," only a subset of predicted TFBSs are occupied by a TF. Identifying those TFBSs can be challenging. One way to address this and provide TFBSs with stronger confidence is to combine experimental data with computational models, such as PWMs. This strategy is employed to compute TFBSs stored in the UniBind database (Figure 3.16) [131, 132].

Current experimental techniques aim to identify TF binding, but in practice, the data reflect direct and indirect or non-specific TF binding [133]. It is challenging to separate the two signals computationally. In addition, while the

3.3. Use of position weight matrices

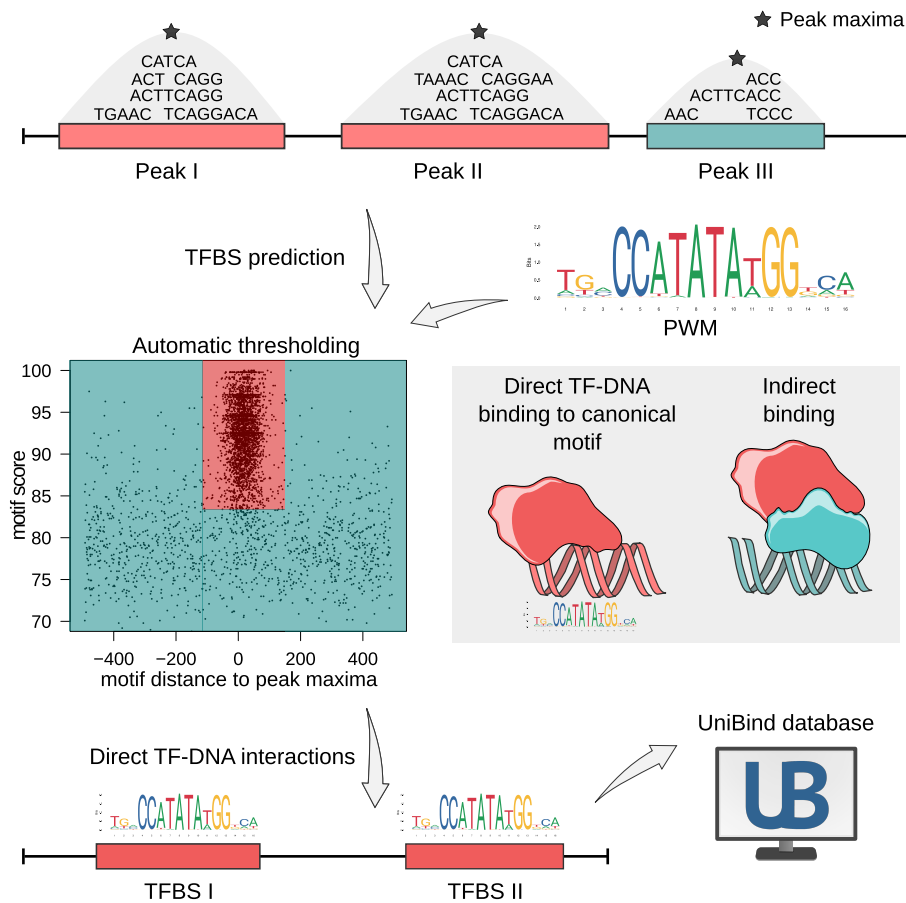


Figure 3.16: An example of combining experimental data with computational models to infer direct TF-DNA interactions. The ChIP-eat pipeline takes ChIP-seq data in peaks and predicts transcription factor binding sites (TFBSs) using the DAMO-trained position weight matrix (PWM) of the profiled TF. The automatic thresholding selects TFBSs, which represent direct TF-DNA binding to the canonical motif (red), from the rest of the predictions, representing indirect or unspecific binding (blue). The final sets of direct TFBSs are collected for each analyzed dataset and compiled into the UniBind database. Adapted from [131].

TFs interact with the DNA through short motifs, the ChIP-seq peaks are usually hundreds of base pairs long. A peak summit defines a location within the peak where the signal (the amount of ChIP-seq reads) is the highest. The ChIP-eat computational approach identifies TFBSs that the TF of interest directly binds. The method combines ChIP-seq peak data and PWMs to produce sets of direct TFBSs (Figure 3.16) [131, 132]. Given a set of peaks and a PWM, the TFBS enrichment zone is detected using an automatic threshold. The enrichment zone contains TFBSs with high PWM scores and close proximity to ChIP-seq peak summits. The resulting set of TFBSs represent direct TF-DNA interactions [131, 132]. The resulting TFBSs are of the size of the DNA binding motif. TFBSs from 9 species are provided to the community as a UniBind database [131]. UniBind TFBSs can be a great starting point for further analysis. For example, enrichment analysis can identify enriched TFBSs in a set of regions of interest and provide information on the transcription regulation driving TFs [131].

3.3.3 Annotating *de novo* discovered motifs

Another use of collections of TF binding profiles is an annotation of *de novo* discovered motifs. Computational tools are designed to compare the motifs of interest against the provided collection of known motifs. For example, TOMTOM compares the query motif against a database of target motifs. It allows for gapped motif-motif alignments, computes the p-value for each motif, and performs a Bonferroni correction to derive motif E value. TOMTOM integrates a variety of metrics to score similarity (for example, Euclidian distance and Pearson's correlation) [134]. RSAT also has a tool, *compare-matrices*, to annotate discovered motifs when the known motif library is provided [112, 114]. The tool can compare two collections of PWMs, and TOMTOM uses different similarity scoring functions. Multiple or all metrics can be compared and later summarized by computing the mean rank [112].

3.3.4 Prediction of TF co-binding events

As with *de novo* motif discovery, predicting TF binding sites is usually limited to individually analyzing each TF binding profile, so deciphering the possible co-occurrence of TF motifs requires specialized tools. The prediction of a co-binding event usually focuses on the prediction of a secondary motif near known TFBSs (primary motif), which could indicate a potential binding of a cooperating TF (Figure 3.17). Out of all sequences with a primary motif, a subset might not have any co-binding events (Figure 3.17A), have a predicted secondary motif with a specific spacer and orientation from the primary motif (Figure 3.17B) or have more than one predictions with specific spacings (Figure 3.17C). Prediction of the secondary motif is as important as determining the motif grammar.

SpaMo is a computational tool that analyzes co-binding events using a library of known motifs to detect possible co-binding TFs relative to the indicated primary TF [135]. The method was developed to analyze ChIP-seq data, where the profiled TF is the primary TF. It summarizes the spacing and orientation

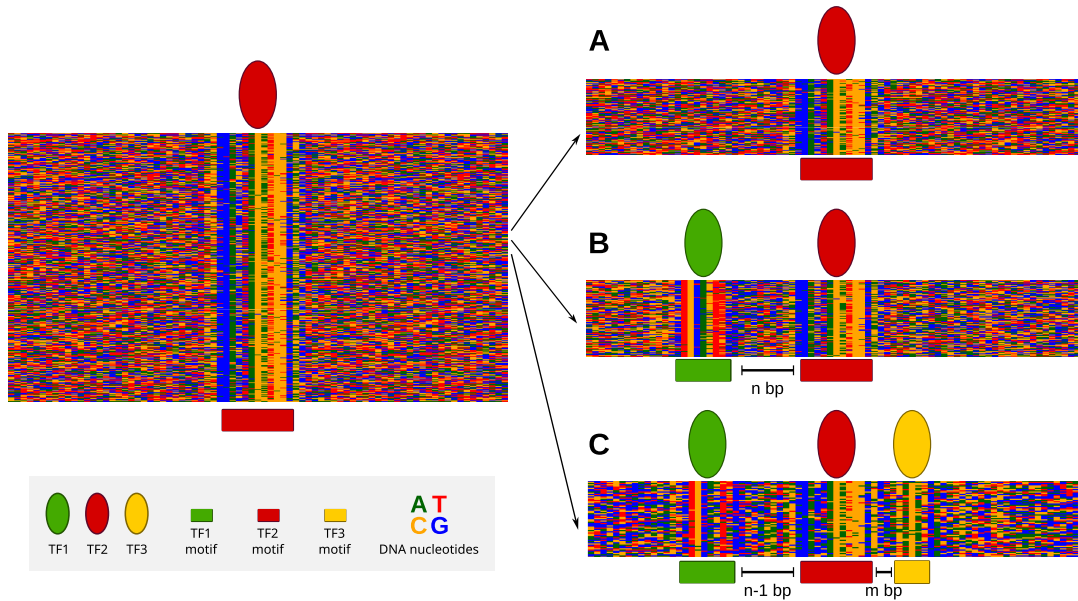


Figure 3.17: Prediction of TF co-binding events. Co-binding events can be investigated by discovering co-binding motifs. A set of genomic sequences with known TFBSs harbor a primary motif bound by a TF2. Some of these sequences do not contain any secondary motif that could indicate a potential co-binding event (**A**). Other sequences can harbor a secondary motif n bp upstream of the primary and potentially bound by a TF1 (**B**) or two motifs: TF1 motif a shorter distance from the primary and TF3 that is m bp downstream of the primary motif (**C**).

between the primary and secondary motifs and provides information on the match's significance. The algorithm tests every secondary motif to determine whether the predicted TFBSs have enriched spacing concerning the primary TFBSs [135]. A significant limitation of SpaMo and similar tools is the requirement of known TF binding motif libraries, meaning previously uncharacterized DNA patterns are missed.

It is crucial to consider the context in which individual TFs are binding. The cooperative binding of multiple TFs in the regulatory regions enables fine-tuning transcription. Therefore, developing computational tools for analyzing such mechanisms would be beneficial in expanding our knowledge about TF cooperativity.

